WILFRID LAURIER UNIVERSITY

Faculty of Science - Department of Physics and Computer Science

CP476A - Internet Computing

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MILESTONE 01

Project Planning and Design

TaskFlow

A Personal Task Management System

Team Members

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1. Project Proposal

1.1 Problem Statement

In today's fast-paced academic and professional environments, individuals struggle to effectively track and manage their tasks, assignments, and daily responsibilities. Many people rely on scattered sticky notes, multiple apps, or memory alone, leading to missed deadlines, forgotten tasks, and increased stress. Students, in particular, face the challenge of balancing coursework, personal commitments, and extracurricular activities without a centralized system to help prioritize and organize their workload.

Existing solutions are often either too complex with unnecessary features that create a steep learning curve, or too simplistic without essential functionality like due dates, priorities, and status tracking. There is a clear need for a straightforward, web-based task management tool that provides core functionality without overwhelming the user.

1.2 Motivation

TaskFlow addresses this gap by providing a clean, intuitive interface for personal task management. The motivation behind this project stems from our own experiences as students managing multiple courses, assignments, and personal responsibilities. We wanted to create a tool that we would actually use ourselves – something simple enough to adopt immediately but powerful enough to genuinely improve productivity.

1.3 Target Users

- **University Students:** Managing coursework, assignments, exams, and personal tasks
- **Young Professionals:** Tracking work tasks, meetings, and personal to-dos
- **Anyone seeking organization:** Individuals who want a simple, no-frills task management solution

1.4 App Concept and Scope

TaskFlow is a web-based personal task management application that allows users to create, organize, track, and complete tasks. The application follows the traditional CRUD (Create,

Read, Update, Delete) workflow while incorporating essential features like task prioritization, due dates, and status tracking.

In Scope:

- User registration and authentication (login/logout)
- Complete CRUD operations for tasks
- Task attributes: title, description, due date, priority level, status
- Task filtering and sorting by various criteria
- Responsive web interface accessible on desktop browsers
- Input validation and basic security measures

Out of Scope:

- Team/collaborative task management
- Mobile native applications
- Calendar integration or external API connections
- Email notifications or reminders
- File attachments or rich media support

1.5 Features List

Priority	Feature	Description
Must Have	User Authentication	Register, login, and logout functionality
Must Have	Create Tasks	Add new tasks with title, description, due date, priority
Must Have	View Tasks	Display all tasks in a list/dashboard view

Must Have	Edit Tasks	Modify task details and update status
Must Have	Delete Tasks	Remove tasks with confirmation prompt
Must Have	Task Status	Track as Pending, In Progress, or Completed
Should Have	Filter Tasks	Filter by status, priority, or due date
Should Have	Sort Tasks	Sort by due date, priority, or creation date
Should Have	Priority Levels	Visual indicators for Low, Medium, High priority
Could Have	Search Tasks	Search by title or description keywords
Could Have	Dashboard Stats	Overview showing task counts by status

2. Requirements (User Stories)

The following user stories define the core requirements for TaskFlow. Each story follows the standard format and includes specific, testable acceptance criteria.

US-1: User Registration

*As a **new user**, I want to **create an account** so that **I can save and access my personal tasks securely.***

Story Points: 5

Priority: High

Sprint Assignment: 1

Status: Not Started

Acceptance Criteria:

- Registration form includes fields for username, email, password, and confirm password
- Username must be unique and 3-20 characters long
- Email must be valid format and unique in the system
- Password must be at least 8 characters with one number and one letter
- Password and confirm password must match
- Appropriate error messages displayed for validation failures
- Upon successful registration, user is redirected to the login page with a success message

US-2: User Login and Logout

*As a **registered user**, I want to **log in and log out** so that **only I can access and manage my tasks.***

Story Points: 3

Priority: High

Sprint Assignment: 1

Status: Not Started

Acceptance Criteria:

- Login form accepts email/username and password
- Invalid credentials display a generic error message (not revealing which field is wrong)
- Successful login redirects to the task dashboard
- User session persists until logout or browser closure

- Logout button is visible on all authenticated pages
- Logout clears session and redirects to login page

US-3: Create a New Task

*As a **logged-in user**, I want to **create a new task** so that **I can track something I need to do.***

Story Points: 2

Priority: High

Sprint Assignment: 1

Status: Not Started

Acceptance Criteria:

- 'Add Task' button is prominently visible on the dashboard
- Task form includes: title (required, max 100 chars), description (optional, max 500 chars), due date (optional, date picker), priority (dropdown: Low/Medium/High, default Medium)
- New tasks are created with status 'Pending' by default
- Form validation prevents submission with empty title
- Due date cannot be set in the past
- Upon successful creation, task appears in the task list immediately
- A confirmation message is displayed after successful creation

US-4: View and Filter Tasks

*As a **logged-in user**, I want to **view all my tasks and filter them** so that **I can focus on relevant tasks and understand my workload.***

Story Points: 2

Priority: Medium

Sprint Assignment: 1

Status: Not Started

Acceptance Criteria:

- Dashboard displays all user's tasks in a list format
- Each task shows: title, due date (if set), priority indicator, and current status
- Filter dropdown allows filtering by status (All/Pending/In Progress/Completed)
- Filter dropdown allows filtering by priority (All/Low/Medium/High)
- Filters can be combined (e.g., 'High priority' + 'Pending')
- If no tasks match filters, display 'No tasks found' message
- Tasks are sorted by due date (soonest first) by default

US-5: Edit an Existing Task

*As a **logged-in user**, I want to **edit my tasks** so that **I can update details or change the status as I make progress.***

Story Points: 1

Priority: Medium

Sprint Assignment: 1

Status: Not Started

Acceptance Criteria:

- Each task in the list has an 'Edit' button/icon
- Clicking Edit opens a form pre-populated with current task data
- All fields (title, description, due date, priority, status) are editable
- Status can be changed to: Pending, In Progress, or Completed
- Same validation rules apply as task creation
- 'Cancel' button discards changes and returns to task list
- Successful update shows confirmation and reflects changes immediately

US-6: Delete a Task

*As a **logged-in user**, I want to **delete tasks I no longer need** so that **my task list stays clean and relevant.***

Story Points: 1

Priority: High

Sprint Assignment: 1

Status: Not Started

Acceptance Criteria:

- Each task has a 'Delete' button/icon
- Clicking Delete shows a confirmation dialog: 'Are you sure you want to delete this task?'
- Confirmation dialog shows task title for clarity
- 'Cancel' dismisses the dialog without deleting
- 'Confirm' permanently removes the task from the database
- Task is immediately removed from the displayed list upon deletion

US-7: Mark Task as Complete (Quick Action)

*As a **logged-in user**, I want to **quickly mark a task as complete** so that **I can update status without opening the edit form.***

Story Points: 1

Priority: High

Sprint Assignment: 1

Status: Not Started

Acceptance Criteria:

- Each non-completed task has a checkbox or 'Complete' button
- Clicking completes the task in one action (no additional confirmation)
- Task status changes to 'Completed' immediately
- Visual indicator shows completed state (e.g., strikethrough, dimmed)

- Completed tasks can be 'uncompleted' via Edit if needed

3. Wireframes

The following wireframes illustrate the primary user workflow and key screens of the TaskFlow application. Navigation flow is shown through numbered annotations.

3.1 User Flow Overview

The primary workflow follows this path: Landing Page → Registration/Login → Task Dashboard → Create/Edit/View Tasks → Logout

Screen 1: Login Page

Purpose: Entry point for registered users to access their account.

- Logo and application name at top
- Email/username input field
- Password input field (with show/hide toggle)
- 'Login' button
- Link to Registration page ('Don't have an account? Sign up')
- Error message area below form

Screen 2: Registration Page

Purpose: Allow new users to create an account.

- Username input field
- Email input field
- Password input field
- Confirm password input field

- 'Register' button
- Link to Login page ('Already have an account? Login')
- Inline validation messages for each field

Screen 3: Task Dashboard (Main Screen)

Purpose: Central hub for viewing and managing all tasks.

- Header: Logo, username display, Logout button
- '+ Add Task' button (prominent)
- Filter controls: Status dropdown, Priority dropdown
- Task list with columns: Checkbox, Title, Due Date, Priority badge, Status badge, Actions (Edit/Delete)
- Empty state message when no tasks exist

Screen 4: Add/Edit Task Form

Purpose: Create new tasks or modify existing ones.

- Modal/overlay or separate page
- Title input (required)
- Description textarea (optional)
- Due date picker
- Priority dropdown (Low/Medium/High)
- Status dropdown (visible in Edit mode only)
- 'Save' and 'Cancel' buttons
- Validation error messages

Screen 5: Delete Confirmation Dialog

Purpose: Prevent accidental task deletion.

- Modal overlay

- Warning icon
- 'Delete Task?' heading
- Task title displayed
- 'Cancel' and 'Delete' buttons

Note: Detailed wireframe mockups are provided as separate image files in the submission.

4. Data Planning

TaskFlow requires a simple relational database structure with two primary entities. The design prioritizes simplicity while supporting all required functionality.

4.1 Key Data Entities

Entity 1: Users

Stores registered user account information.

Attribute	Type	Notes
user_id	INT	Primary Key, Auto-increment
username	VARCHAR(20)	Unique, Not Null
email	VARCHAR(100)	Unique, Not Null
password_hash	VARCHAR(255)	Hashed password, Not Null

created_at	DATETIME	Account creation timestamp

Entity 2: Tasks

Stores all task records with their attributes.

Attribute	Type	Notes
task_id	INT	Primary Key, Auto-increment
user_id	INT	Foreign Key → Users, Not Null
title	VARCHAR(100)	Not Null
description	TEXT	Optional, max 500 chars
due_date	DATE	Optional
priority	ENUM	'low', 'medium', 'high' - Default 'medium'
status	ENUM	'pending', 'in_progress', 'completed'
created_at	DATETIME	Task creation timestamp
updated_at	DATETIME	Last modification timestamp

4.2 Relationships

Users → Tasks: One-to-Many relationship. One user can have many tasks; each task belongs to exactly one user. This is enforced through the foreign key constraint on tasks.user_id referencing users.user_id.

When a user is deleted, their associated tasks should also be deleted (CASCADE delete) to maintain referential integrity and prevent orphaned records.

5. Team Plan

5.1 Team Roles and Responsibilities

Team Member	Primary Role	Responsibilities
Allen Dolgonos	Team Lead / Front-End	Project coordination, UI design, HTML/CSS/JS implementation, wireframes
Chris Ooi En Shen	Back-End / Database	Server-side logic, API endpoints, database design and queries, security
Joey Sufaj	Full-Stack / Testing	Integration, testing, documentation, code review support

Note: Roles will rotate for learning purposes. All members will participate in code reviews and documentation.

5.2 Communication Plan

- **Primary Communication:** Discord server (text channels + voice for meetings)
- **Quick Updates:** Discord #general channel for daily check-ins

- **Code Discussion:** GitHub Pull Requests and Issues
- **Document Sharing:** GitHub repository Wiki

5.3 Meeting Cadence

- **Weekly Team Meeting:** Tuesdays, 6:00 PM - 7:00 PM (Discord voice)
- **Ad-hoc Sessions:** As needed for pair programming or problem-solving
- **Pre-submission Review:** Day before each milestone deadline

5.4 Definition of Done

A task is considered 'Done' when ALL of the following criteria are met:

1. **Code Complete:** Feature is fully implemented and functional
2. **Tested:** Developer has tested the feature manually; edge cases considered
3. **Code Reviewed:** At least one other team member has reviewed the code
4. **Documented:** Code has appropriate comments; README updated if needed
5. **Merged:** Code is merged into main branch without conflicts
6. **Kanban Updated:** Task card moved to 'Done' column with completion notes

6. GitHub Setup Evidence

6.1 Repository Information

- **Repository Name:** cp476-taskflow
- **URL:** https://github.com/josefsufaj/cp476-taskflow
- **Visibility:** Public
- **License:** MIT

6.2 README Structure

The repository README includes the following sections:

- Project title, description, and team members
- Technology stack (to be finalized)
- Setup instructions (placeholder for Milestone 02)
- Project structure overview
- Links to Wiki and Project Board

6.3 Kanban Board Setup

GitHub Projects board has been configured with the required columns:

Column	Purpose
Backlog	All identified tasks not yet scheduled for current sprint
Ready	Tasks that are defined, estimated, and ready to be worked on
In Progress	Tasks currently being worked on by a team member
In Review	Completed work awaiting code review or testing
Done	Completed tasks that meet the Definition of Done

6.4 Initial Tasks Created

The following tasks have been created and assigned for Milestone 01:

- [M01] Write project proposal - Allen (Done)

- [M01] Create user stories - Joey (Done)
- [M01] Design wireframes - Allen (Done)
- [M01] Plan database schema - Chris (Done)
- [M01] Set up GitHub repo and board - Joey (Done)
- [M01] Initialize Wiki with meeting minutes - Chris (Done)
- [M01] Compile and review final submission - All (Done)

7. Activity Blog / Wiki

The team maintains an Activity Blog in the GitHub Wiki. Below is the initial planning entry documenting our first team meeting and task assignments.

Meeting Minutes - January 28, 2026

Date: Tuesday, January 28, 2026, 6:00 PM - 7:15 PM

Location: Discord Voice Channel

Attendees: Allen Dolgonos, Chris Ooi En Shen, Joey Sufaj

Meeting Type: Project Kickoff / Planning

Agenda Items Discussed

7. **Project Idea Selection:** Team reviewed the suggested project topics from the course outline. Unanimous agreement to build a Task/To-Do Management System due to clear CRUD workflow, manageable scope, and practical utility.

8. **Scope Definition:** Discussed what features are essential vs. nice-to-have. Agreed on MoSCoW prioritization. Decided against team collaboration features to keep scope manageable.

9. **Technology Stack:** Will use technologies covered in CP476. Front-end: HTML, CSS, vanilla JavaScript. Back-end: PHP or Node.js (to confirm with instructor). Database: MySQL.

10. **Role Assignment:** Allen volunteered to lead and handle the front-end. Chris has database experience, will handle back-end. Joey will bridge both and lead testing.

11. **GitHub Setup:** Joey will create the repository tonight. All members to be added as collaborators.

Decisions Made

- Project name: 'TaskFlow'
- Meeting cadence: Weekly on Tuesdays at 6 PM
- Communication: Discord for daily updates, GitHub for code discussion
- Deadline for M01 tasks: January 30, 2026 (day of submission)

Task Assignments

Task	Assignee	Due Date
Write project proposal (Section 1)	Allen	Jan 30
Create user stories with acceptance criteria	Joey	Jan 30
Design wireframes for all screens	Allen	Jan 30
Plan database schema	Chris	Jan 30
Set up GitHub repo, Projects board	Joey	Jan 30
Initialize Wiki, document meeting	Chris	Jan 30
Review and compile final report	All	Jan 30

Action Items for Next Meeting

- All: Review each other's completed sections by Jan 30
- Allen: Prepare wireframe walkthrough for team review
- Chris: Confirm technology stack with instructor if needed
- Joey: Ensure all Kanban tasks are up to date

--- End of Milestone 01 Report ---